

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 May/June 2021 Worked Solutions

May/June 2021 paper rebuilt with the worked solution after each question.

9

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P1**

PAST PAPER

WMA11/01 May/June 2021

May/June 2021 | 9 questions | 75 marks

9

questions

75

marks

Answers

worked solution
after each
question

Question 1

Differentiation

1. The curve C has equation

$$y = \frac{x^2}{3} + \frac{4}{\sqrt{x}} + \frac{8}{3x} - 5 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(4)

The point $P(4,3)$ lies on C .

- (b) Find the equation of the normal to C at the point P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)

(Total 8 marks)

Worked Solution - Question 1

1. Rewrite using powers

$$y = \frac{x^2}{3} + 4x^{-1/2} + \frac{8}{3}x^{-1} - 5.$$

2. Differentiate term by term

$$\frac{dy}{dx} = \frac{2x}{3} - 2x^{-3/2} - \frac{8}{3}x^{-2} = \frac{2x}{3} - \frac{2}{x\sqrt{x}} - \frac{8}{3x^2}.$$

3. Find the tangent gradient at P

$$\text{At } x = 4, \frac{dy}{dx} = \frac{8}{3} - \frac{1}{4} - \frac{1}{6} = \frac{9}{4}.$$

4. Find the normal gradient

The normal gradient is the negative reciprocal of $\frac{9}{4}$, so $m_n = -\frac{4}{9}$.

5. Write the normal equation

Using $P(4, 3)$, $y - 3 = -\frac{4}{9}(x - 4)$. Multiplying by 9 gives $9y - 27 = -4x + 16$, hence $4x + 9y - 43 = 0$.

Final answer

$$(a) \frac{dy}{dx} = \frac{2x}{3} - \frac{2}{x\sqrt{x}} - \frac{8}{3x^2}.$$

$$(b) 4x + 9y - 43 = 0.$$

Question 2

Polynomials

2. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = ax^3 + (6a + 8)x^2 - a^2x$$

where a is a positive constant.

Given $f(-1) = 32$

(a) (i) show that the only possible value for a is 3

(ii) Using $a = 3$ solve the equation

$$f(x) = 0 \quad (5)$$

(b) Hence find all real solutions of

(i) $3y + 26y^{\frac{2}{3}} - 9y^{\frac{1}{3}} = 0$

(ii) $3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0 \quad (5)$

(Total 10 marks)

Worked Solution - Question 2

1. Use $f(-1) = 32$

$f(-1) = -a + (6a + 8) + a^2 = a^2 + 5a + 8$. Since $f(-1) = 32$,
 $a^2 + 5a - 24 = 0$.

2. Choose the positive value of a

$a^2 + 5a - 24 = (a - 3)(a + 8) = 0$. Since a is positive, the only possible value is $a = 3$.

3. Solve $f(x) = 0$

Using $a = 3$,

$$f(x) = 3x^3 + 26x^2 - 9x = x(3x^2 + 26x - 9) = x(3x - 1)(x + 9).$$

4. List the roots

$$x = 0, x = \frac{1}{3}, \text{ or } x = -9.$$

5. Solve the equation in y

Let $u = y^{1/3}$. Then $y = u^3$ and $y^{2/3} = u^2$, so $3y + 26y^{2/3} - 9y^{1/3} = 0$ becomes $3u^3 + 26u^2 - 9u = 0$.

6. Convert back to y

From part (a), $u = -9, 0, \frac{1}{3}$. Therefore $y = u^3$, giving $y = -729, 0, \frac{1}{27}$.

7. Solve the exponential equation

Let $u = 9^z$. Then $3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0$ becomes $3u^3 + 26u^2 - 9u = 0$.

8. Use positivity of 9^z

Since $9^z > 0$, the only valid root is $u = \frac{1}{3}$. Hence $9^z = \frac{1}{3}$, so $(3^2)^z = 3^{-1}$ and $2z = -1$. Therefore $z = -\frac{1}{2}$.

Final answer

$$(a)(i) a = 3. \quad (a)(ii) x = -9, 0, \frac{1}{3}.$$

$$(b)(i) y = -729, 0, \frac{1}{27}.$$

$$(b)(ii) z = -\frac{1}{2}.$$

Question 3

Basic Trigonometry

3. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

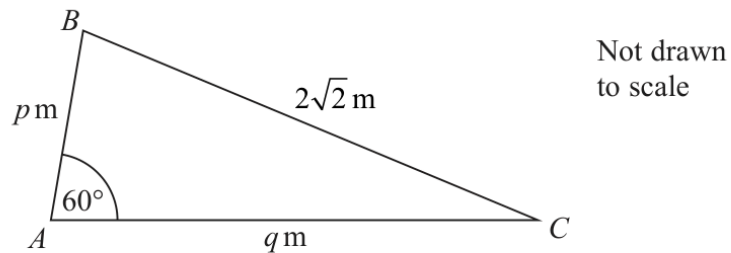


Figure 1

Figure 1 shows the plan view of a flower bed.
The flowerbed is in the shape of a triangle ABC with

- $AB = p$ metres
- $AC = q$ metres
- $BC = 2\sqrt{2}$ metres
- angle $BAC = 60^\circ$

- (a) Show that

$$p^2 + q^2 - pq = 8 \quad (2)$$

Given that side AC is 2 metres longer than side AB , use algebra to find

- (b) (i) the exact value of p ,
(ii) the exact value of q . (5)

Using the answers to part (b),

- (c) calculate the exact area of the flower bed. (2)

(Total 9 marks)

Worked Solution - Question 3

1. Apply the cosine rule

$BC^2 = AB^2 + AC^2 - 2(AB)(AC) \cos 60^\circ$. Since $BC = 2\sqrt{2}$, $AB = p$ and $AC = q$, this gives $8 = p^2 + q^2 - 2pq \left(\frac{1}{2}\right)$.

2. Show the required result

Therefore $p^2 + q^2 - pq = 8$.

3. Use $q = p + 2$

Side AC is 2 metres longer than AB , so $q = p + 2$. Substitute into $p^2 + q^2 - pq = 8$.

4. Solve for p

$p^2 + (p + 2)^2 - p(p + 2) = 8$, so $p^2 + 2p - 4 = 0$. Hence $p = -1 \pm \sqrt{5}$. Since $p > 0$, $p = \sqrt{5} - 1$.

5. Find q

$q = p + 2 = \sqrt{5} + 1$.

6. Find the exact area

Area = $\frac{1}{2}pq \sin 60^\circ$. Now $pq = (\sqrt{5} - 1)(\sqrt{5} + 1) = 4$, so area
 $= \frac{1}{2}(4) \left(\frac{\sqrt{3}}{2}\right) = \sqrt{3} \text{ m}^2$.

Final answer

(a) $p^2 + q^2 - pq = 8$.

(b) $p = \sqrt{5} - 1$, $q = \sqrt{5} + 1$.

(c) $\sqrt{3} \text{ m}^2$.

Question 4**Integration**

4. Find

$$\int \frac{(3\sqrt{x} + 2)(x - 5)}{4\sqrt{x}} dx$$

writing each term in simplest form.

(6)

(Total 6 marks)

Worked Solution - Question 4

1. Expand the numerator

$$(3\sqrt{x} + 2)(x - 5) = 3x\sqrt{x} - 15\sqrt{x} + 2x - 10.$$

2. Divide by $4\sqrt{x}$

$$\frac{(3\sqrt{x} + 2)(x - 5)}{4\sqrt{x}} = \frac{3x}{4} - \frac{15}{4} + \frac{1}{2}x^{1/2} - \frac{5}{2}x^{-1/2}.$$

3. Integrate term by term

$$\int \left(\frac{3x}{4} - \frac{15}{4} + \frac{1}{2}x^{1/2} - \frac{5}{2}x^{-1/2} \right) dx = \frac{3x^2}{8} - \frac{15x}{4} + \frac{1}{3}x^{3/2} - 5x^{1/2} + c.$$

4. Write each term simply

So the integral is $\frac{3x^2}{8} - \frac{15x}{4} + \frac{x\sqrt{x}}{3} - 5\sqrt{x} + c.$

Final answer

$$\frac{3x^2}{8} - \frac{15x}{4} + \frac{x\sqrt{x}}{3} - 5\sqrt{x} + c.$$

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8 marks

Question 5

Graphs of Functions

5.

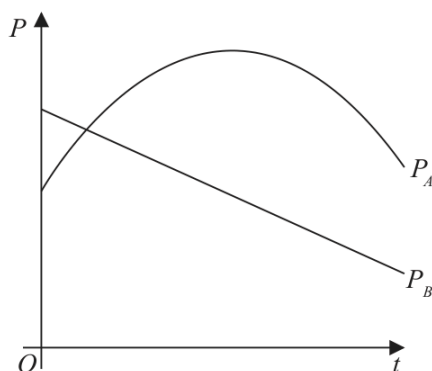


Figure 2

The share value of two companies, company *A* and company *B*, has been monitored over a 15-year period.

The share value P_A of **company A**, in millions of pounds, is modelled by the equation

$$P_A = 53 - 0.4(t - 8)^2 \quad t \geq 0$$

where t is the number of years after monitoring began.

The share value P_B of **company B**, in millions of pounds, is modelled by the equation

$$P_B = -1.6t + 44.2 \quad t \geq 0$$

where t is the number of years after monitoring began.

Figure 2 shows a graph of both models.

Use the equations of one or both models to answer parts (a) to (d).

- (a) Find the difference between the share value of **company A** and the share value of **company B** at the point monitoring began. (2)
- (b) State the maximum share value of **company A** during the 15-year period. (1)
- (c) Find, using algebra and showing your working, the times during this 15-year period when the share value of **company A** was greater than the share value of **company B**. (4)
- (d) Explain why the model for **company A** should not be used to predict its share value when $t = 20$ (1)

(Total 8 marks)

Worked Solution - Question 5

1. Evaluate both models at $t = 0$

$$P_A(0) = 53 - 0.4(0 - 8)^2 = 53 - 25.6 = 27.4. \text{ Also}$$

$$P_B(0) = -1.6(0) + 44.2 = 44.2.$$

2. Find the initial difference

The difference is $44.2 - 27.4 = 16.8$, so the share values differ by **16.8** million pounds.

3. State the maximum of company A

$P_A = 53 - 0.4(t - 8)^2$ has maximum value **53**, reached when $t = 8$, which is within the 15-year period.

4. Set up the inequality for company A greater than company B

$$53 - 0.4(t - 8)^2 > -1.6t + 44.2.$$

5. Rearrange to a quadratic inequality

Expanding gives $-0.4t^2 + 8t - 16.8 > 0$. Multiplying by -2.5 reverses the inequality: $t^2 - 20t + 42 < 0$.

6. Find the crossing times

$$t = \frac{20 \pm \sqrt{400 - 168}}{2} = 10 \pm \sqrt{58}. \text{ Therefore company A is greater between the two roots.}$$

7. Restrict to the 15-year period

Since the model is monitored over $0 \leq t \leq 15$, the required times are $10 - \sqrt{58}$

8. Explain the limitation

The model should not be used at $t = 20$ because it was only built from a 15-year monitoring period; 20 years is outside that range.

Final answer

(a) 16.8 million pounds.

(b) 53 million pounds.

(c) $10\sqrt{58}$

(d) $t = 20$ is outside the 15-year model.

Question 6

Integration

6. The curve C has equation $y = f(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$

- $f'(x) = \frac{32}{3x^2} + 3 - 2(\sqrt[3]{x})$

- (a) find the equation of the tangent to C at P . Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

- (b) Find, in simplest form, $f(x)$.

(5)

(Total 8 marks)

Worked Solution - Question 6

1. Find the gradient at P

$$f'(x) = \frac{32}{3x^2} + 3 - 2\sqrt[3]{x}. \text{ At } x = 8,$$

$$f'(8) = \frac{32}{3(64)} + 3 - 2(2) = \frac{1}{6} - 1 = -\frac{5}{6}.$$

2. Write the tangent equation

Using $P(8, 2)$, $y - 2 = -\frac{5}{6}(x - 8)$, hence $y = -\frac{5}{6}x + \frac{26}{3}$.

3. Integrate $f'(x)$

$$f'(x) = \frac{32}{3}x^{-2} + 3 - 2x^{1/3}, \text{ so } f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + c.$$

4. Use $P(8, 2)$ to find c

$$2 = -\frac{32}{24} + 24 - \frac{3}{2}(16) + c = -\frac{4}{3} + c, \text{ so } c = \frac{10}{3}.$$

5. Write $f(x)$

$$f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + \frac{10}{3}.$$

Final answer

$$(a) y = -\frac{5}{6}x + \frac{26}{3}.$$

$$(b) f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + \frac{10}{3}.$$

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10 marks

Question 7

Radians

7.

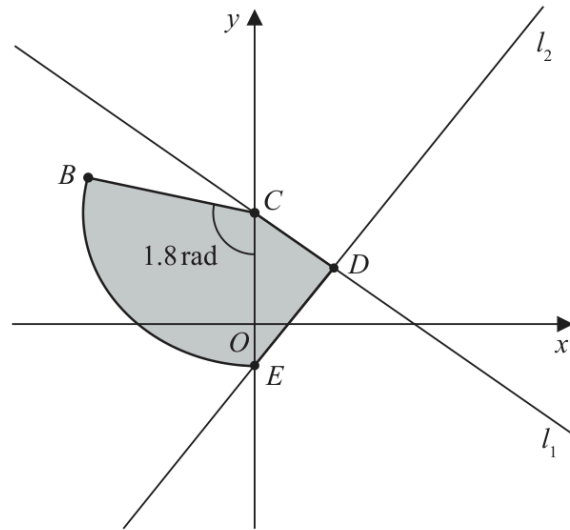


Figure 3

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y -axis at the point C , as shown in Figure 3.

(a) State the y coordinate of C .

(1)

The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y -axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\frac{14}{3}$

(3)

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

(c) find the length of arc BE .

(3)

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE .

(d) Calculate the exact area of the region $CBED$.

(3)

(Total 10 marks)

Worked Solution - Question 7

1. Find C

On the y-axis, $x = 0$. From $4y + 3x = 48$, $4y = 48$, so the y-coordinate of C is 12.

2. Find the gradient of l_1

$4y + 3x = 48$ gives $y = -\frac{3}{4}x + 12$, so the gradient of l_1 is $-\frac{3}{4}$.

3. Use perpendicular gradients for l_2

Since l_2 is perpendicular to l_1 , its gradient is $\frac{4}{3}$. Through $D(8, 6)$,
 $y - 6 = \frac{4}{3}(x - 8)$.

4. Show the y-coordinate of E

$y = \frac{4}{3}x - \frac{32}{3} + 6 = \frac{4}{3}x - \frac{14}{3}$. Therefore when $x = 0$, $E_y = -\frac{14}{3}$.

5. Find the radius of the sector

$C = (0, 12)$ and $E = \left(0, -\frac{14}{3}\right)$, so $CE = 12 + \frac{14}{3} = \frac{50}{3}$.

6. Find arc BE

For sector BCE , $r = \frac{50}{3}$ and $\theta = 1.8 = \frac{9}{5}$. Hence arc
 $BE = r\theta = \frac{50}{3} \cdot \frac{9}{5} = 30$.

7. Find the sector area

Sector area = $\frac{1}{2}r^2\theta = \frac{1}{2}\left(\frac{50}{3}\right)^2\left(\frac{9}{5}\right) = 250$.

8. Find the triangle area and total area

Triangle CDE has vertical base $CE = \frac{50}{3}$ and horizontal height 8, so its area is $\frac{1}{2} \cdot \frac{50}{3} \cdot 8 = \frac{200}{3}$. Total area = $250 + \frac{200}{3} = \frac{950}{3}$.

Final answer

(a) 12.

(b) $E_y = -\frac{14}{3}$.

(c) 30.

(d) $\frac{950}{3}$.

Question 8

Polynomials

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

- (a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point P is the minimum point of C_1

- (b) Deduce the coordinates of P .

(1)

A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A , B , C and D are constants.

Given that C_2

- passes through P
- intersects the x -axis at -4 , -2 and 3

- (c) find, making your method clear, the values of A , B , C and D .

(5)

(Total 9 marks)

Worked Solution - Question 8

1. Complete the square

$$3x^2 + 6x + 9 = 3(x^2 + 2x) + 9 = 3((x + 1)^2 - 1) + 9 = 3(x + 1)^2 + 6.$$

2. Deduce the minimum point

The minimum occurs when $(x + 1)^2 = 0$, so $x = -1$ and $y = 6$. Therefore $P = (-1, 6)$.

3. Use the x-intercepts of C_2

Since C_2 intersects the x-axis at -4 , -2 and 3 , it has form $y = k(x + 4)(x + 2)(x - 3)$.

4. Use point P to find k

Substitute $P = (-1, 6)$: $6 = k(3)(1)(-4) = -12k$, so $k = -\frac{1}{2}$.

5. Expand the cubic

$$(x + 4)(x + 2)(x - 3) = (x^2 + 6x + 8)(x - 3) = x^3 + 3x^2 - 10x - 24.$$

6. Read off A, B, C and D

Thus $y = -\frac{1}{2}x^3 - \frac{3}{2}x^2 + 5x + 12$, so $A = -\frac{1}{2}$, $B = -\frac{3}{2}$, $C = 5$ and $D = 12$.

Final answer

(a) $3(x + 1)^2 + 6$.

(b) $P = (-1, 6)$.

(c) $A = -\frac{1}{2}$, $B = -\frac{3}{2}$, $C = 5$, $D = 12$.

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7 marks

Question 9

Trigonometric Functions

9.

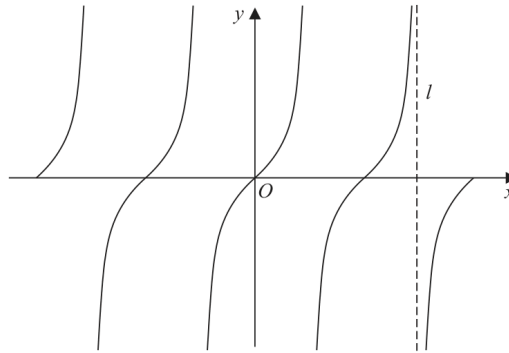


Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

The line l , shown in Figure 4, is an asymptote to $y = \tan x$

(a) State an equation for l .

(1)

A copy of Figure 4, labelled Diagram 1, is shown on the next page.

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \quad -2\pi \leq x \leq 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leq x \leq 2\pi$

(4)

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leq x \leq 40\pi$

(ii) $-10\pi \leq x \leq \frac{5}{2}\pi$

(2)

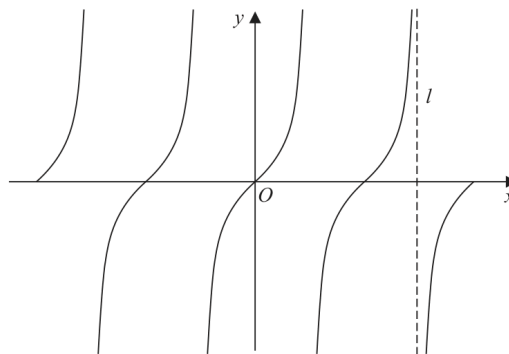


Diagram 1

(Total 7 marks)

Worked Solution - Question 9

1. State the shown tangent asymptote

The shown dashed asymptote is the right-hand positive asymptote before 2π , so l has equation $x = \frac{3\pi}{2}$.

2. Describe the reciprocal graph

For $y = \frac{1}{x} + 1$, the vertical asymptote is $x = 0$ and the horizontal asymptote is $y = 1$. The right branch lies above $y = 1$ and the left branch lies below $y = 1$.

3. Count intersections on -2π to 2π

The number of solutions of $\tan x = \frac{1}{x} + 1$ is the number of intersections of the two graphs. From the sketch on $-2\pi \leq x \leq 2\pi$, there are 5 intersections.

4. Count solutions for $0 \leq x < 40\pi$

From the repeated tangent branches and the reciprocal curve, there is one intersection in each positive interval between consecutive tangent asymptotes. From 0 to 40π , this gives 40 solutions.

5. Count solutions for $-10\pi \leq x \leq 5\pi/2$

Extending the same intersection pattern across the stated interval gives 14 intersections, so there are 14 solutions.

Final answer

(a) $x = \frac{3\pi}{2}$.

(b)(i) $y = 1$ is the horizontal asymptote.

(b)(ii) 5.

(c)(i) 40.

(c)(ii) 14.